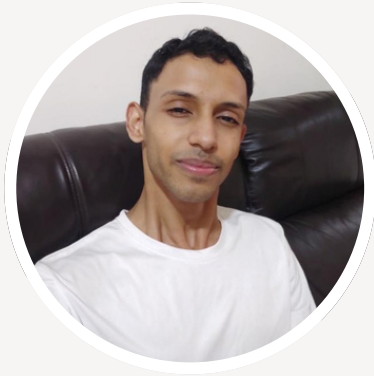




PRANAV SAGAR

Full Stack AI Engineer

ABOUT ME

A detail-oriented Full Stack AI Developer specializing in backend development with Python and FastAPI. Proficient in building secure, scalable RESTful APIs featuring JWT authentication and robust data validation using Pydantic. Experienced in designing and managing both relational (PostgreSQL, MySQL, SQLite3) and non-relational (MongoDB) databases. Adept in modern DevOps practices, including Docker containerization and deployment across various cloud platforms like Google Cloud and Vercel. Actively expanding expertise into advanced AI/ML frameworks like LangGraph and Hugging Face Transformers, alongside full-stack technologies including React.js and Next.js.

EXPERTISE

Backend & API Development
Frontend Web Development
DevOps & Deployment
Databases
AI & Machine Learning
Robotics & Simulation
Leadership & Calisthenics Coach

LANGUAGE

English ————— Native

CONTACTS



[LinkedIn](#)



pranavsagar529@gmail.com



+91 - 9110638699

PROJECTS

Full-Stack Banking Application (In progress)

Developing a feature-complete banking application designed to handle real-world transactions and manage persistent user data.

Full-Stack Screen Recording & Video Sharing Platform (In progress)

Building a platform that allows users to record their screen and share the video with others with AI rich features.

AI Voice-Agent for Robotic Arm

Implemented a WebRTC-based pipeline for low-latency transfer of video and audio data, enabling the arm to respond to user's visual and vocal queues.

Secured first place in an ISTE (Indian Society for Technical Engineers) competition at New Horizon's College of Engineering with the same.

Finger-Movement Controlled Robotic Arm

Designed and implemented a novel control algorithm based on Google's own pre-trained Machine learning models and a custom geometric algorithm orchestrated with OpenCV to translate finger coordinates into precise robotic arm commands.



TECHNICAL SKILLS

Backend & API Development

Python (Advanced Asynchronous Programming), FastAPI, RESTful API Design, JWT Authentication, Pydantic Data Validation Docker Fundamentals such as Port Mapping, Docker Compose, Dockerfile, Docker Hub

Frontend Web Development (Learning)

Next.js 16, React.js 19, React Native, Tailwind CSS, Three.js, GSAP

Databases

Relational: PostgreSQL, MySQL, SQLite3
Non-Relational: MongoDB

DevOps & Deployment

Docker Core Fundamentals, Containerization, Google Cloud Platform (GCP, Cloud Run), Vercel, Hostinger

AI & Machine Learning

Frameworks: LangChain, LangGraph
Concepts: RAG Agents, MCP, Knowledge Graphs (Neo4j), Model Fine-Tuning (Hugging Face Transformers)

Robotics & Simulation:

NVIDIA Isaac SIM, CoppeliaSIM, OpenCV, WebRTC based AI Voice Agents

EDUCATION

M.S. Ramaiah University of Applied Sciences, Peenya, Bangalore Bachelor of Engineering in Robotics Engineering Highest CGPA: 9.81	2022-2026
--	-----------

RESEARCH & DEVELOPMENT

Cursor AI Clone Building a voice-based AI Agent Designed to replicate Cursor	2025
Scalable RAG Agents Architecting and developing RAG agents utilizing asynchronous queues and distributed workers for enhanced performance.	2025
Complex AI Workflows Leveraging LangGraph and LangChain with a MongoDB backend to create sophisticated AI agentic workflows	2025
AI Model Fine-Tuning Customizing and fine-tuning models from the Hugging Face ecosystem for specialized use-cases.	2025
Full-Stack Clones Replicating the functionality and design of popular applications like Google Drive and Uber, and creating clones of Awwwards-winning 3D websites.	2026

LEADERSHIP & ROBOTICS EXPERIENCE

ISRO Robotics Challenges - URSC 2025 Team Lead Selected by the university Dean to lead the official team for a national-level robotics competition.	2025
IIT Bombay's E-yantra Robotics Competition 2025 Team Lead Leading a team in one of India's most prestigious robotics competitions.	2025
Robotics Club Technical Team Lead Organized and led foundational workshops on Python, robotics, and AI for over 30 first and second-year students and Actively mentoring students, providing technical guidance for their projects.	2025
University Guest Speaker Invited to be a guest speaker and resource person at Dayananda Sagar Engineering College and Tasked with educating 5th and 6th-semester students on the integration of AI and Simulation in modern robotics.	2026
Class Representative Coordinating, communicating, and overseeing a variety of daily activities while also shaping future academic schedules in collaboration with the faculty.	2022-2026

EXTRACURRICULAR ACTIVITIES & ACHIEVEMENTS

Founder & Coach, The Calisthenics Club of MS RUAS	2025
Successfully founded and gained official university approval for the first-ever calisthenics club on campus.	
Research, Planning and Designing Roadmaps	2026
Personally design and implement training programs for dozens of student members, leading to measurable skill progression.	
Strengths and Achievements	2025
Achieved elite-level calisthenics skills, including front, back levers, straddle planche, and heavy weighted pull-ups, pushups (25, 50 kgs) and dips (50 kgs).	
Intercollegiate Activities	2026
Took Part in several inter-collegiate activities such as SIH(Smart-India Hackathon) Ideathon-2025 & National Technology Day 2024	

PASSIONS & ASPIRATIONS

Beyond my technical pursuits, I am a dedicated researcher and a voracious reader, constantly exploring advancements in humanoid robotics and biotechnology. My core interest lies in applying principles of biomechanics—not only to create novel and effective calisthenics workouts but also to inform the design of bio-inspired robots. This passion fuels my entrepreneurial vision to one day launch two distinct ventures:

Neuro-Force: A biotech robotics company aimed at replicating complex human biomechanics in machines and bringing them to life with advanced AI.

SCARF: A hybrid fitness startup built on the principles of Absolute Strength, Calisthenics, Athleticism, Relative Health, and Movement Freedom.

This drive for excellence extends to my athletic life, where I have competed in state-level badminton tournaments and won first place in multiple swimming competitions.

Certifications and References

- FastAPI - The Complete Course 2025
- Modern AI - The Complete Overview (In Progress)
- Full Stack Developer Certification (In Progress)
- Other Certification Links

Adithya S Bhat
Working in Accenture
Phone: +91 - 9686896228
Email : adityabhat52@gmail.com

Malatesh K
CEO, Vision Technologies
Phone: +91 - 9448481224
Email : malateshk1402@gmail.com

Sreenath Janardhanan
Deputy General Manager,
TATA Consulting Engineers Ltd.,
Phone: +91 - 8861384354
Email : sjanardhanan@tataconsultingengineers.com